PIMPRI CHINCHWAD EDUCATION TRUST'S

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Data Science for Engineers

Data Collection, Preprocessing and EDA of
Heart Disease Dataset

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1. Introduction

Data science is an interdisciplinary field that focuses on extracting meaningful insights and knowledge from data using statistical methods, programming, and machine learning techniques. With the rapid growth of digital data in healthcare, finance, and other industries, analyzing and understanding data has become increasingly important for decision making.

However, raw data collected from different sources is often incomplete, inconsistent, or noisy. Therefore, before applying any analytical or machine learning techniques, the data must undergo data preprocessing. Data preprocessing involves tasks such as handling missing values, transforming variables, and scaling features so that the dataset becomes suitable for analysis.

Another important step in the data science workflow is Exploratory Data Analysis (EDA). EDA helps in understanding the structure of the dataset by analyzing distributions, relationships between variables, and patterns within the data. Visualization techniques such as histograms, boxplots, and correlation heatmaps are commonly used to identify trends and potential relationships among features.

In this study, the Heart Disease dataset from the UCI Machine Learning Repository is used to demonstrate the process of data collection, preprocessing, and exploratory data analysis. The goal is to understand the dataset, clean and prepare the data, and analyze key attributes that may influence the presence of heart disease.

2. Problem Statement

Heart disease is one of the leading causes of mortality worldwide. Early detection and analysis of risk factors can help in better diagnosis and prevention. Medical datasets containing patient information such as age, cholesterol level, blood pressure, and other clinical attributes can be analyzed to understand patterns related to heart disease.

The objective of this study is to analyze the Heart Disease dataset obtained from the UCI Machine Learning Repository and perform data preprocessing and exploratory data analysis. The dataset will be cleaned by handling missing values and applying feature scaling techniques. Exploratory data analysis will then be performed to identify patterns, distributions, and relationships between the medical attributes and the presence of heart disease.

3. Dataset Description

The dataset used in this study is the UCI Heart Disease Dataset, obtained from the UCI Machine Learning Repository. The dataset contains medical information of patients that can be used to analyze factors related to the presence of heart disease.

The dataset consists of 303 instances (records) and 14 attributes, including patient demographic information, clinical measurements, and diagnostic results. The attributes include both numerical and categorical variables. The target variable indicates whether a patient has heart disease or not.

During preprocessing, the dataset was cleaned by handling missing values and transforming the target variable into a binary classification problem where 0 represents absence of heart disease and 1 represents presence of heart disease.

Figure 3.1 : Shows first 5 records of the dataset

```
df.head()
```

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	target
0	63.0	1.0	1.0	145.0	233.0	1.0	2.0	150.0	0.0	2.3	3.0	0.0	6.0	0
1	67.0	1.0	4.0	160.0	286.0	0.0	2.0	108.0	1.0	1.5	2.0	3.0	3.0	2
2	67.0	1.0	4.0	120.0	229.0	0.0	2.0	129.0	1.0	2.6	2.0	2.0	7.0	1
3	37.0	1.0	3.0	130.0	250.0	0.0	0.0	187.0	0.0	3.5	3.0	0.0	3.0	0
4	41.0	0.0	2.0	130.0	204.0	0.0	2.0	172.0	0.0	1.4	1.0	0.0	3.0	0

Table 3.1 : Shows attributes of the dataset

Attribute	**Description**
age	Age of the patient
sex	Gender of the patient (1 = male, 0 = female)
cp	Chest pain type
trestbps	Resting blood pressure
chol	Serum cholesterol level
fbs	Fasting blood sugar
restecg	Resting electrocardiographic results

thalach	Maximum heart rate achieved
exang	Exercise heart rate achieved
oldpeak	ST depression induced by exercise
slope	Slope of peak exercise ST segment
ca	Number of major vessels colored by fluoroscopy
thal	Thalassemia defect type
target	Presence of heart disease

4. Data Collection

The dataset used in this study was obtained from the UCI Heart Disease Dataset, which is available on the UCI Machine Learning Repository. This repository provides publicly available datasets that are widely used for research and educational purposes in machine learning and data science.

The heart disease dataset contains medical records collected from patients at the Cleveland Clinic Foundation. It includes several clinical attributes such as age, cholesterol level, blood pressure, and electrocardiographic results that are useful for analyzing the presence of heart disease.

For this analysis, the processed Cleveland dataset containing 303 patient records and 14 attributes was used. The dataset was downloaded and imported into Google Colab, where Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn were used for data preprocessing and exploratory data analysis.

5. Data Preprocessing

Data preprocessing is an important step in data analysis, as real-world datasets often contain missing values, inconsistencies, and variables with different scales. Proper preprocessing ensures that the dataset becomes clean, consistent, and suitable for further analysis. In this study, several preprocessing techniques were applied to the heart disease dataset before performing exploratory data analysis.

Handling Missing Values

During the initial inspection of the dataset, missing values were found in the attributes ca and thal. In the dataset, these missing values were represented by the symbol “?”.

To address this issue, the missing values were replaced using appropriate imputation techniques.

- Median imputation was applied to the ca attribute because it is a numerical variable and the median is less affected by extreme values.
- Mode imputation was applied to the thal attribute since it represents categorical information, and the mode represents the most frequently occurring category.

These techniques ensured that the dataset no longer contained missing values while preserving the overall distribution of the data.

Figure 5.1 : Shows missing values before and after the preprocessing

age	0
sex	0
cp	0
trestbps	0
chol	0
fbs	0
restecg	0
thalach	0
exang	0
oldpeak	0
slope	0
ca	4
thal	2
target	0

dtype: int64

age	0
sex	0
cp	0
trestbps	0
chol	0
fbs	0
restecg	0
thalach	0
exang	0
oldpeak	0
slope	0
ca	0
thal	0
target	0

dtype: int64

Target Variable Transformation

The original dataset contained target values ranging from 0 to 4, where:

- 0 indicated absence of heart disease
- 1 to 4 indicated different levels of heart disease

For simplicity and clarity, the target variable was converted into a binary classification problem:

- 0 → No heart disease
- 1 → Presence of heart disease

This transformation makes the dataset easier to analyze and is commonly used in medical classification problems.

Feature Scaling

The dataset contains several numerical attributes such as age, cholesterol level, blood pressure, and heart rate, which have different value ranges. To ensure that all features contribute equally during analysis, feature scaling was applied.

In this study, Standardization (Z-score scaling) was used through the StandardScaler technique. This method transforms the data using the formula:

$$\square = (\square - \square)/\square$$

where:

- $\square$ is the original value
- $\square$ is the mean of the feature
- $\square$ is the standard deviation

After standardization, the features have a mean close to 0 and a standard deviation close to 1, which helps in improving the performance of analytical and machine learning methods.

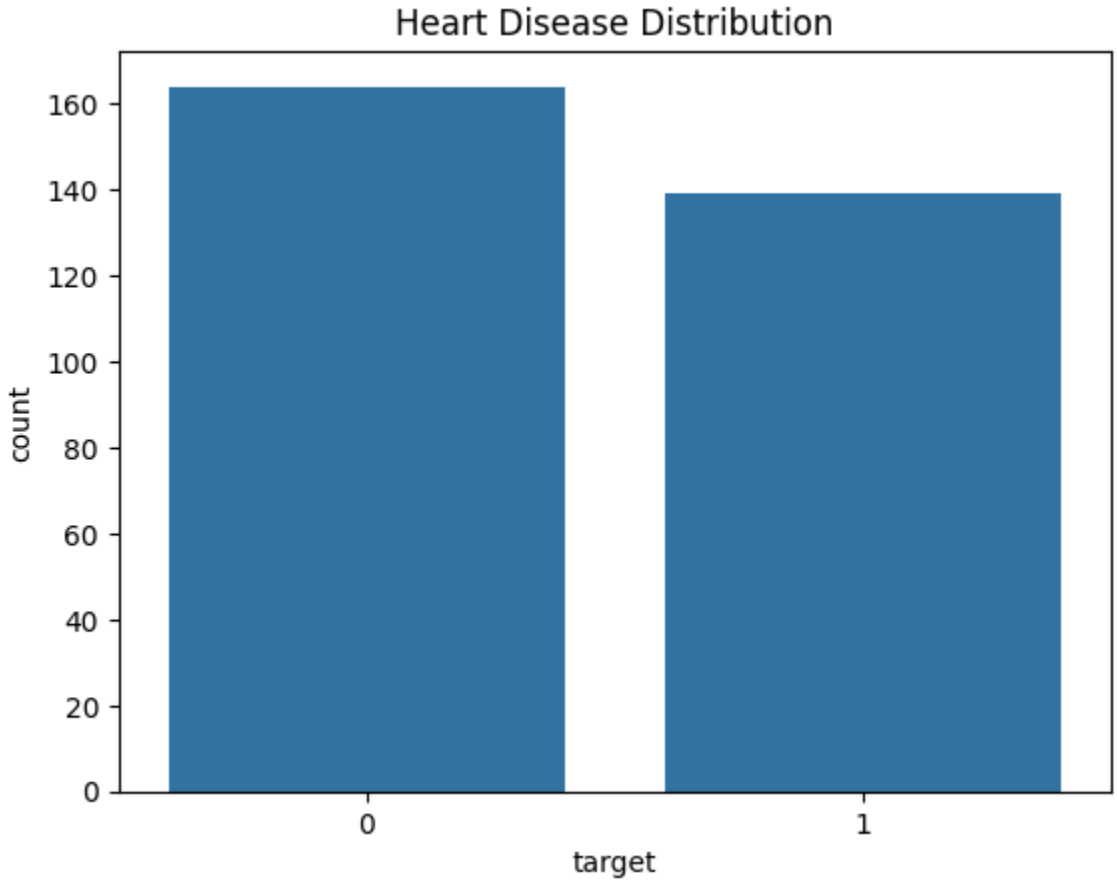
6. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is used to understand the structure and characteristics of the dataset. It helps identify patterns, relationships between variables, and possible trends in the data. Visualization techniques such as histograms, box plots, and correlation heatmaps were used to analyze the heart disease dataset.

Distribution of Heart Disease Cases

A count plot was used to visualize the distribution of patients with and without heart disease as shown in figure 6.1.
The analysis shows that the dataset contains patients both with and without heart disease, providing a balanced dataset suitable for classification analysis. Understanding the distribution of the target variable helps in identifying potential class imbalance problems and evaluating the dataset before model development.

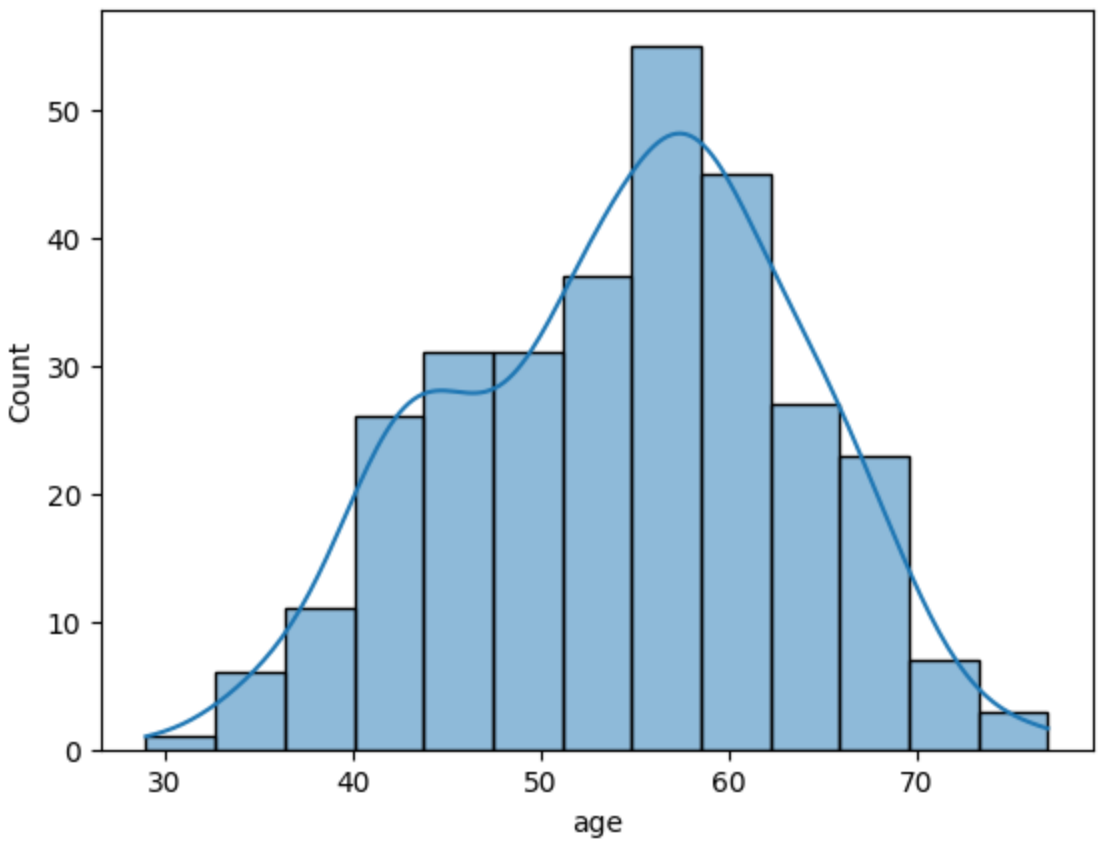
Figure 6.1 : Shows distribution of heart disease patients (without - 0, with - 1)



Age Distribution of Patients

A histogram was used to examine the distribution of patient ages in the dataset as shown in figure 6.2.
The visualization shows that most patients fall within the 40–60 year age range, which indicates that heart disease risk increases in middle-aged and older individuals. The distribution also shows fewer cases among very young patients.

Figure 6.2 : Shows age distribution of patients

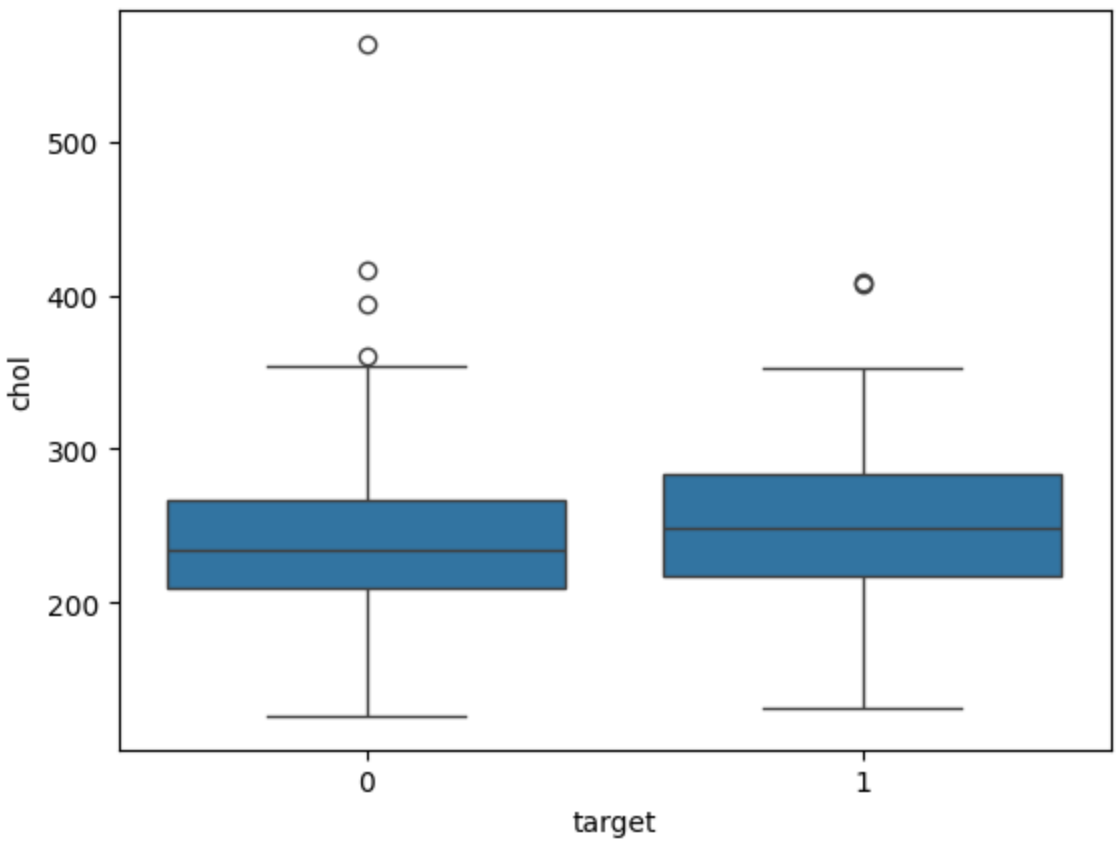


Cholesterol Level vs Heart Disease

A boxplot was used to analyze the relationship between cholesterol levels and the presence of heart disease as shown in figure 6.3.

The visualization suggests that patients with heart disease tend to have relatively higher cholesterol levels compared to those without heart disease. This indicates that cholesterol may be an important factor influencing heart disease risk.

Figure 6.3 : Shows cholesterol vs heart disease

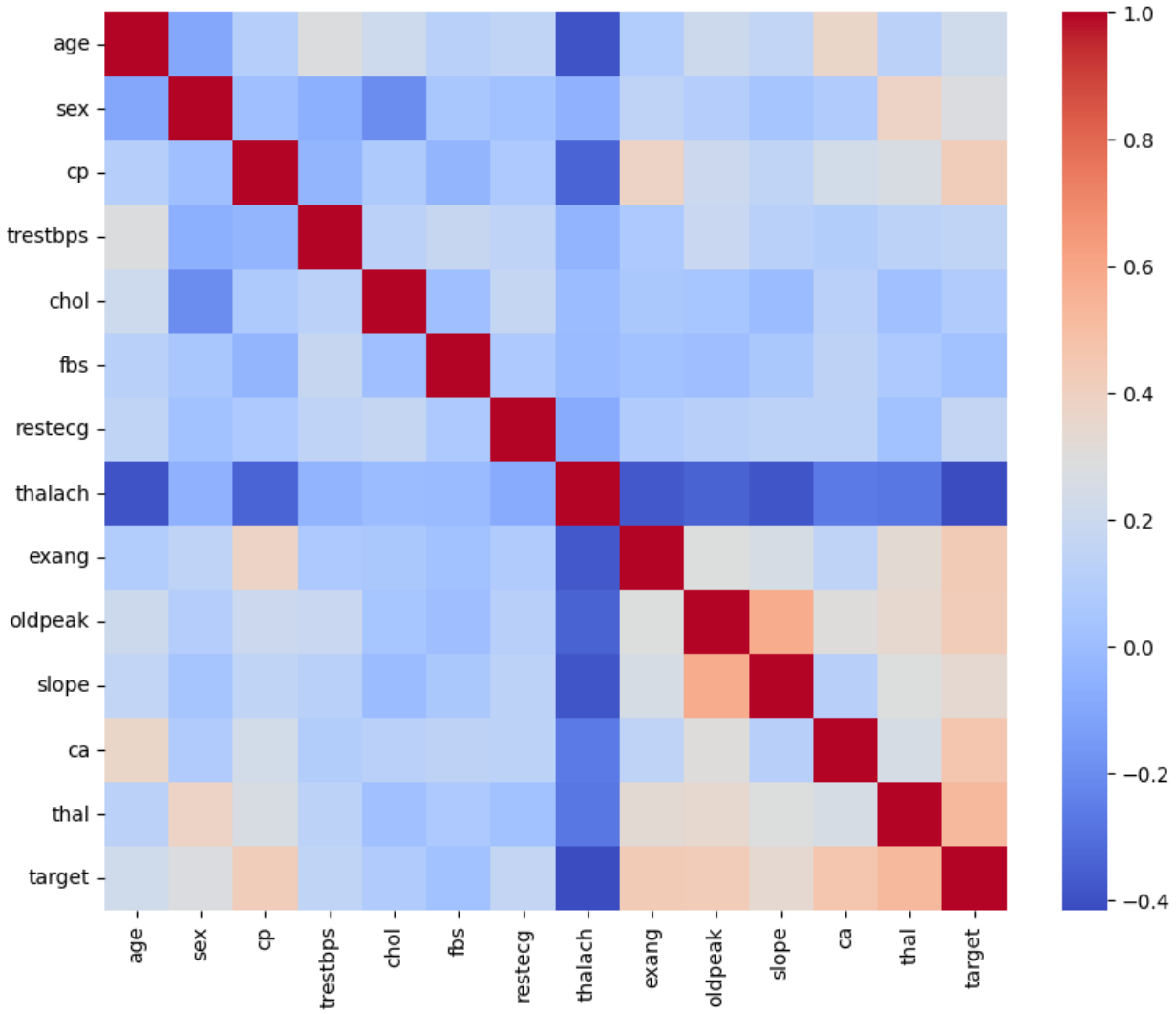


Correlation Analysis

A correlation heatmap was generated to analyze the relationships between different attributes in the dataset.

The heatmap shows the strength of relationships between variables using color intensity. Some attributes show moderate correlation with the target variable, indicating their potential importance in predicting heart disease. Correlation analysis helps in identifying features that may significantly influence the outcome.

Figure 6.4 : Shows correlation heatmap of dataset features



7. Planning and Design

The analysis of the heart disease dataset followed a structured workflow to ensure that the data was properly prepared and analyzed. The overall process involved several steps starting from data collection to exploratory analysis.

First, the dataset was obtained from the UCI Heart Disease Dataset available on the UCI Machine Learning Repository. The dataset was then imported into the Google Colab environment where Python libraries were used for analysis.

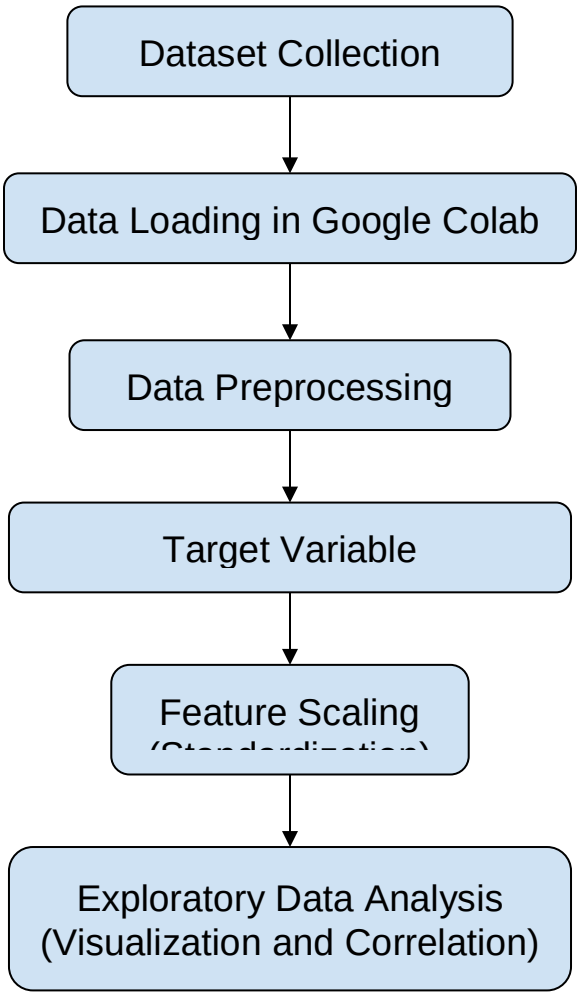
The next step involved data preprocessing, where the dataset was inspected for missing values and inconsistencies. Missing values in certain attributes were handled using appropriate imputation

techniques such as median and mode replacement. The target variable was also transformed into a binary format to simplify the analysis.

After cleaning the dataset, feature scaling was applied to standardize numerical attributes so that all variables would have comparable scales. This step ensures that features with larger numerical ranges do not dominate the analysis.

Finally, Exploratory Data Analysis (EDA) was performed to understand the relationships between different attributes and the target variable. Visualization techniques such as histograms, box plots, and correlation heatmaps were used to identify patterns and trends within the dataset.

Workflow Diagram



8. Tools and Technologies Used

The analysis of the heart disease dataset was performed using the following tools and technologies:

- **Google Colab** was used as the development environment for executing Python code and performing data analysis. It provides a cloud-based platform that allows running Python notebooks without requiring local installation.

- **Python** was used as the primary programming language due to its extensive support for data analysis and machine learning libraries.

- The **Pandas** library was used for data manipulation and preprocessing tasks such as loading the dataset, handling missing values, and transforming variables.

- **NumPy** was used for numerical computations and efficient handling of array-based data.

- **Matplotlib** was used to create visualizations such as histograms and other graphical plots for analyzing the data.

- **Seaborn** was used to generate advanced visualizations such as count plots, boxplots, and correlation heatmaps.

- The **Scikit-learn** library was used to perform feature scaling using the StandardScaler technique.

9. Conclusion

In this study, the Heart Disease dataset was analyzed to understand the process of data collection, preprocessing, and exploratory data analysis. The dataset was obtained from the UCI Heart Disease Dataset and imported into the Google Colab environment for analysis.

Several preprocessing steps were performed to prepare the dataset for analysis. Missing values in the dataset were handled using appropriate imputation techniques such as median and mode replacement. The target variable was transformed into a binary format to simplify the classification problem. Feature scaling was also applied using standardization to ensure that all numerical attributes had comparable ranges.

Exploratory Data Analysis was conducted using visualization techniques such as histograms, boxplots, count plots, and correlation heatmaps. These visualizations helped in understanding the

distribution of the data and identifying relationships between different medical attributes and the presence of heart disease.

Overall, the preprocessing and exploratory analysis provided valuable insights into the dataset and prepared it for further machine learning model development. Proper data preparation and analysis are essential steps in any data science workflow to ensure accurate and reliable results.

10. References

UCI Machine Learning Repository – Heart Disease Dataset

Python Documentation – https://www.python.org

Scikit-learn Documentation – https://scikit-learn.org

Pandas Documentation – https://pandas.pydata.org